XT FIRMWARE FOR LXR-02

VERSION 1.2.13 · SUPPLEMENT TO THE LXR OWNERS MANUAL

SAMPLES & WAVES

31 WAV samples, 8 single-cycle waves and 4 wavetables from the SD card, with start, length and loops.

SEQUENCER

A step grid per track, ratchets and roll locks, trigger tracks, LFO start, lock trigger and instant pattern switching.

EFFECTS

A second FX slot, sends, pre-fader option and a master bus on St1: saturation, compressor and EQ.

ENGINES & MORE

Modal, FM2 and wavetable (test), kit and voice morph, cue, WAV recorder, INIT templates.

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github.com/shellbeat-jpg/lxr-desktop-sample
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1 > NOTES

This supplement describes what the **XT firmware 1.2.13** (formerly OE02-Samples+) adds to the LXR-02 and its manual.

It contains the modifications of **OpenElixir-02 1.2** (Moritz Waser), marked OE.

1.1. ABOUT THE TABLES

As in the manual: **display name** (three letters), full name, function. The edit view (press the encoder) shows the full name and the value.

Buttons by their label below the step button, e.g. **SHIFT + FX** (step 15).

1.2. WHAT IS STORED WHERE

Global: kept only after SAVE → Global. All new settings with their storage: appendix A.

1.3. INSTALLATION

1. **LXR_XT_V1.2.13.img** into the top folder of the card; delete all other **LXR*.img** there.
2. Switch on with the encoder button held until "LXR Updater" appears.
3. "Update OK!", then delete the image.
4. The update erases the samples: leave **SAMPLES** on the card, the next start imports them again.

After switching off, wait ≈ 10 s.

1.4. COMPATIBILITY

Kits and patterns of the original firmware and of OpenElixir-02 load; new settings start with default values.

New kits also load in older firmwares (without the additions); voices with a sample or wave stay silent there, new engines play the default engine.

Going back: install the image of OpenElixir-02 or Erica Synths the same way.

2 > MEMORY MANAGEMENT

2.1. SD CARD

Besides the projects (and **REC**, chapter 5.2):

Folder / file	Contents
SAMPLES	one-shot samples, WAV
SAMPLES/LOOPS	loop samples, WAV
SAMPLES/WAVES	up to 8 single-cycle waves, WAV
SAMPLES/TABLES	up to 4 wavetables, WAV (3.14)
BIGFLASH	in **SAMPLES**: 1.5 MB memory (2.3)
INIT.SND	INIT template: top folder (global), project folder (2.5)

2.2. SAMPLE IMPORT

At start the LXR compares the folders with the internal flash (names and sizes) and imports again on changes. The samples also play without a card.

Rule	
Format	WAV, 16-bit PCM, mono or stereo (mixed to mono), any sample rate
Names	8.3 format (**01KICK.WAV**); the slots **S1** … **S31** follow the order of the names, numbers by value (**2KICK** before **10KICK**)
Count	up to 31 samples (both folders together) and 8 waves
Length	about 5.9 s mono in total at 44.1 kHz (11.9 s at 22.05 kHz); a stereo file counts at half its size
Ignored	names starting with **.** or **_**

A file replaced by one with the same name and size is not imported again.

START SCREEN

S:12 W:3 T:2 = stored samples, waves and wavetables.

Below: **Used 4.2/5.9s** = used and available sample time.

2.3. EXPERIMENTAL: 1.5 MB SAMPLE MEMORY

The chip reports 1 MB of flash but usually has more.
With **BIGFLASH** in the **SAMPLES** folder: **17.8 s** instead of 5.9 s.

- Not guaranteed (ST). Every sample is verified; on an error "Big flash failed / using 512K" and import into the normal memory.
- If the start hangs permanently: remove **BIGFLASH** or start without a card. Firmware and bootloader are not affected.

2.4. KITS

Additionally in the kit: per voice the engine OE, start and length of the sample, choke group, sends; per kit the order of the FX slots and FX3.

2.5. INIT TEMPLATES FOR VOICES

Hold **LOAD** + step button of the voice (1–7), one step back from kit 0: **INIT: Project** / **Global** / **Factory** shows the source. Click:

Action	Description
Load INIT	from the project template, else the global one, else the engine defaults
Save to Project	voice into the project template
Save to Global	voice into the global template
Del Project INIT, Del Global INIT	remove the voice from the template (click again)

Each complete voice (engine, sound, LFO, velocity, sample). Project template: only in saved projects.

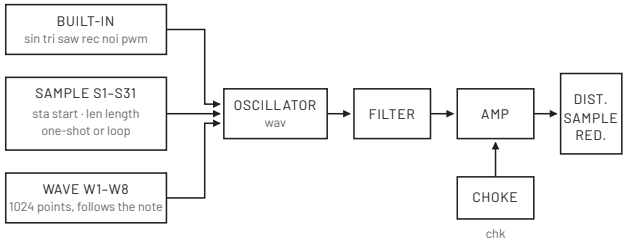
3 > VOICES

3.1. ENGINES PER VOICE OE

Each of the 6 slots selects its engine: drum, snare, cymbal, hi-hat.
New: **modal**, **fm2** and **wavetable** (3.12–3.14).
SHIFT + KIT: engine menu of the active voice. In the kit.

3.2. OSCILLATOR SOURCES

Main oscillator: built-in waveforms, **samples** (own pitch, once or as a loop) or **waves** (single cycles, follow the note).



3.3. OSCILLATOR PAGE (OSC)

Display	Name	Description
wav	Waveform	**sin** ... **pwm** (standard) **S1** ... **S31** (samples) **W1** ... **W8** (waves) Edit view: file name, **L** = loop, slot
sta*	Start	ms, 0 = beginning (all voices, replaces **pwm**)
len*	Length	ms, 0 = to the end (drum 1–3)
coa	Coarse tune	with a sample: 60 = original pitch
oct	Octave rnd	random octave per trigger: **±1**, **+1**, **–1** ... **±3**, **+3**, **–3** (0 included), **off**; step value: 4.3

*: samples only; taken at the trigger, also with LFO/velocity.

FM page of the drum engine: samples and waves also as FM source.

3.4. SAMPLES AND LOOPS

- **SAMPLES**: once per trigger; the AEG decay can end it earlier.
- **SAMPLES/LOOPS**: looped from **sta** to **len**. Each trigger restarts at **sta**; in between the loop runs until the decay fades it out (**dec** = 127: until the next trigger).
- Sequencer stop fades out all loops.

3.5. WAVES

- Up to 8 WAV files in **SAMPLES/WAVES**, one cycle each: **W1** … **W8** (W1 = first name). Stereo: left channel.
- Against clicks: start and end at a zero crossing, rising at both ends, no DC offset.

3.6. MODULATED SAMPLE SELECTION

An LFO or velocity with target **wav** selects among the following samples (or waves):

- Range: from the selected sample, as many more as LFO **mod** / velocity **amt** (S5, amount 4 = S5 … S9). Chosen at the trigger.
- **Velocity layers**: soft to loud versions, named in sequence, select the softest, **amt** = layers – 1.
- **Round robin**: LFO **dst** = **wav**, LFO waveform **cyc**: one sample per trigger.
- **Random**: LFO waveform **rnd**, rate faster than the triggers.
- Only the target **wav** selects samples, not **fwv** / **cwv**.

3.7. AMPLITUDE ENVELOPE (AEG)

Display	Name	Description
chk	Choke	choke group of the track: **off**, **A**, **B**, **C**; edit view: the voice buttons move the tracks to the next group, e.g. **AA-B-CC**
sna	FX1 send	send into slot A, 0 … 127
snb	FX2 send	send into slot B, 0 … 127

- **Choke**: a trigger cuts the other tracks of its group. Tracks 6 and 7 cut each other anyway.
- **Sends**: after pan, after the fader (unless **pre** is on, chapter 6.3). Send/return: slot at 100 % wet.

3.8. LFO PAGE (LFO)

wav	New waveforms
cyc	step counter: one step per trigger, **mod** = count (round robin)
xtr	exponential triangle
si1 tr1 su1 sd1 sq1	one-shot: sine, triangle, saw up/down, square
xu1 xd1 xt1	one-shot exponential: up, down, triangle
rn1	new random value per trigger
w1 … w8 / w1. … w8.	waves (running / one-shot)

Display	Name	Description
ofs	Offset	one-shot: delay until the start (64 ≈ half a cycle)
mod	Amount	edit view: range of the target, e.g. **21..127**
dst	Destination	also rate, amount … of **any** other LFO

One-shots: one cycle per trigger (voice of **rtg**, else its own), then hold. Length: **frq** / **snc**.

3.9. TRACK 7

Tracks 6 and 7 play one voice. Track 7 has its own values for decay, **wav**, **coa** (±24 semitones to track 6), **sna**, **snb**, **pan** and its own LFO. Edit view: "Trk7 …", **=t6** = as track 6. Sends and pan: those of the track that triggered last.

3.10. LFO 2 (TEST)

- Second LFO screen, press LFO again: **LFO 2** of the voice, same parameters.
- Track 7: the LFO page is its own LFO; acts while track 7 plays.
- Stored in the kit, not morphed.

3.11. RANDOM VOICE

- **SHIFT + encoder click** on a voice page (OSC, AEG, MOD, FM, transient, filter; not in the edit view): new random variant of the active voice.
- Amount: **rnd** (PERF, default 32). Switching values change with this probability.
- Kept: volume, pan, **out**, sends, LFO, choke, engine.
- Undo: reload the voice.

3.12. ENGINE MODAL (TEST)

Physical modelling: 6 resonators ring in the modes of a struck body. Drum pages:

Display	Name	Description
dmp	Damping	MOD: decay time, 30 ms … 6 s
inh	Inharmonic	MOD: stiffness, spreads the upper modes
pos	Strike pos.	MOD: edge (0) … centre (127)
hrd	Hardness	FM: soft … hard mallet
mat	Material	FM: 0 = metal … 127 = wood (highs decay faster)
mdl	Model	FM: **mem** membrane, **bar** bar, **str** string, **bel** bell, **plt** plate, **gls** glass
exc	Exciter	FM: **mal** mallet, **nse** noise

The transient sample excites in addition. A new strike adds to the ringing. New voice: decay 110.

3.13. ENGINE FM2 (TEST)

2-operator FM: sine carrier, sine modulator with its own index decay. FM page:

Display	Name	Description
rat	Ratio	modulator : carrier, **.50** … **11** (odd values = metallic)
idx	FM index	modulation depth at the strike
idc	Index decay	5 ms … 4 s
fbk	Feedback	modulator onto itself, towards noise

MOD page: pitch EG as on the drum.

3.14. ENGINE WAVETABLE (TEST)

Scans a set of waves (frames). FM page:

Display	Name	Description
set	Table set	**bas** sine › triangle › square › saw, **usr** from the waves, **all** = both, **t1** … **t4** from **SAMPLES/TABLES**. Edit view: file name or **none>bas**
pos	Wave pos.	0 = first frame … 127 = last
swp	Pos. sweep	the pitch EG pushes the position up at the strike, back to **pos**
scn	Scan mode	**smo** smooth, **stp** stepped, **pp** back and forth, **pps** back and forth stepped

- **Position** = **pos** + **swp** × pitch EG (MOD page: decay, slope; amount 0 = no pitch change) + LFO/automation on **pos**. Not the amp envelope.

OWN SETS: SAMPLES/TABLES

- Up to 4 WAV files (**t1** = first name): 16/24 bit or float, mono/stereo (left).
- Cycle length: from the **clm** chunk (Serum, Vital), else 2048 or 256 (WaveEdit).
- Max. 64 frames; fewer: in-between waves are added. 64 frames ≈ 0.9 s of sample memory, after the samples.

4 > SEQUENCER

4.1. LOCK TRIGGER: VELOCITY 0

Velocity 0: no trigger, no MIDI note; **P1**/**P2** still act ("trigless lock"). SHIFT + encoder click on the automation page clears both locks.

4.2. STEP FLAM: RATCHETS AND ROLL LOCKS

flm	Effect (after x2 … x8)
x12, x24	12 / 24 repeats in the step
x4+, x4-	4 repeats, velocity rising / falling (25 … 100 %)
1/8 1/4 1/2 1/1	roll lock: a repeat every 2, 4, 8, 16 steps up to the next active step

4.3. STEP ACCENT AND OCTAVE

- **acc** (first step page, after **prb**): accent on/off for the step, stored in the pattern.
- **aca** (CONFIG, second page, global): amount +0.1 … +6 dB, default 64 ≈ +3.5 dB.
- Acts like the voice's fader, only for this trigger.
- **oct** (after **sft**): random octave instead of the voice's (3.3): **vce** (as the voice), off, ±1, ±2, ±3, +1, −1.

4.4. INSTANT PATTERN SWITCH

ips on (CONFIG, second page): a pattern chosen by hand starts at the next step at the same position. Chains and songs still switch at the end.

4.5. STEP GRID PER TRACK

SHIFT + LAST STEP: **[16] 1/16 1** = length, grid, step length (1/16 = 1). Press the encoder: edit the grid. ALL mode (LAST STEP again): all tracks.

Grid	Step length · steps per bar
1/4 · 1/4T · 1/8	4 · 8/3 · 2 — 4 · 6 · 8
1/16. · 1/8T · 5/64	6/4 · 4/3 · 5/4 — 10⅔ · 12 · 12⅘
1/16	1 — 16 (default)
1/32. · 1/16T	3/4 · 2/3 — 21⅓ · 24
1/32 · 1/32T	1/2 · 1/3 — 32 · 48

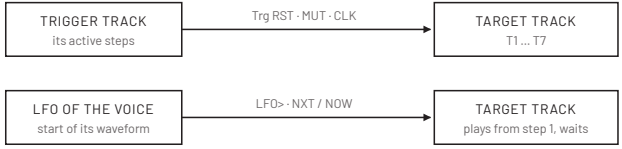
- Each track loops in its length (polymeter); the pattern ends after the longest.
- **rol** and flams count in the track's steps; live recording in 1/16.
- Stored in the pattern.

4.6. TRIGGER TRACKS

LAST STEP page, each encoder click moves on: length › **Start** › grid › **Trg [OFF]** › target **>[T3]** › **LFO>[T3]** › **NXT**/**NOW**. Taken targets are skipped.

Trg	Each active step …
RST	sets the target to step 1 (plays at once)
MUT	mutes the target for the step's duration
CLK	advances the target one step (only with these pulses)

All velocities count; velocity 0 does not sound itself. A track is the target of one source at most.



4.7. LFO START

LFO>[T3]: each start of the LFO waveform of this track's voice (new cycle, **rtg**, one-shot) starts track 3 at step 1: **NXT** at the next step in the target's grid, **NOW** at once.

- The target plays to its last step and waits; silent until the first start.
- With **snc** in time; free running it drifts (generative).
- Trigger and LFO target possible at the same time.

4.8. START, LOOP, GATES, COPYING (TEST)

- **Start** (LAST STEP page): the track plays start … last step; RST and LFO start begin there.
- **Loop** per track: **SHIFT + PROJECT** on/off for the active track. Hold a step, press a second (also on another page); second before first = backwards. Single press: off; ends with a pattern change.
- **Gate**: hold a step, press a later one: the first holds to the end of the second (AEG full, MIDI note off at the end).
- **Prepared**: hold the voice button + step to mark; while off it flashes on the beat.
- **Copy**: hold **COPY**, select steps, release; the next step press pastes them from there with their distances (to the track length).
- **SHIFT + MORPH**: duplicate the range start … last step behind it (length × 2, max. 64).

5 > PERFORMANCE MODE

5.1. PERFORMANCE MENU

Display	Name	Description
rol	Roll	new **24**, **32**, **48** (1/16T, 1/32, 1/32T); in the track's steps
dj res	DJ filter OE	LP/HP and resonance on the final mix
sat frq	Saturation, split	from FX3 (chapter 6.5)
vmv vmo vsw	Voice morph	chapter 5.4
osx	OSR exclude	**on**: voice ignores the global **srt**/**0sr**; edit view: voice buttons
cmx cvl	Cue mix, level	chapter 5.5
rnd	Random	amount of the random voice (3.11)
rec rst	Record WAV	page 3, chapter 5.2

5.2. WAV RECORDER

Records St1 (after FX3, DJ filter, limiter) as WAV to the card: **REC/REC001.WAV** ..., 16-bit stereo, 44.1 kHz.

Display	Name	Description
rec	Record WAV	length: **x1**, **x2**, **x4** passes of the longest track, **end** until stop. Display **arm** / **rec**
rst	Rec start	**ply** with PLAY, **pat** at the next pattern start (also with the sequencer running)

- **Encoder click** in the edit view: arm (creates the file) / cancel / end the recording.
- Ends after the passes, at STOP or on a click: "Saved REC003 1:32".
- **rec** and **rst** are global and stay set after the recording.
- Slow card: dropouts are counted ("2 gaps"). Avoid loading/saving while recording.

5.3. KIT MORPH

- MIDI CC 1 (global channel) sets **mrp**.
- **mrp** 0 → 127: from the active kit to the morph kit. Edit view + **SHIFT**: **Morph>** and target value; **SHIFT + encoder click**: parameter does not morph.
- **mrv** (CONFIG, page 2): morph per voice; edit view: voice buttons.

5.4. VOICE MORPH

Display	Name	Description
vmv	Voices	pair source > target: OFF, 1>2 … 6>5
vmo	Amount	0 = source, 127 = values of the target
vsw	Switch	from this **vmo** on: switching values (waveform, filter type) of the target

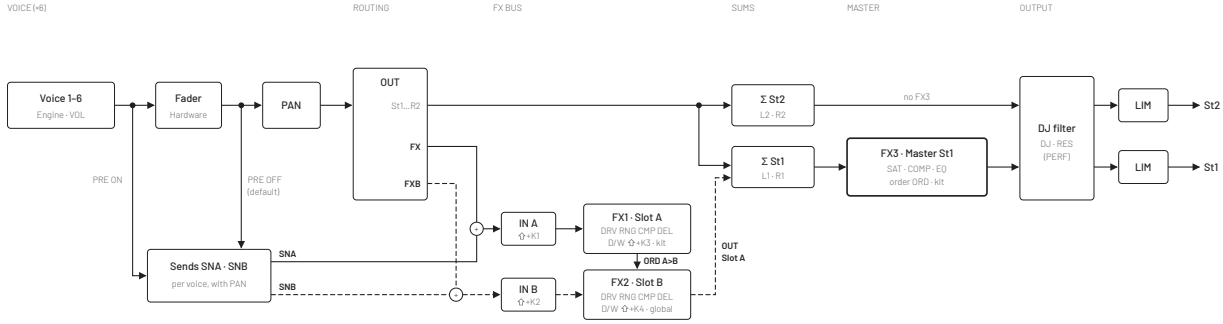
Only **vsw** is stored (global).

5.5. CUE ON ST2

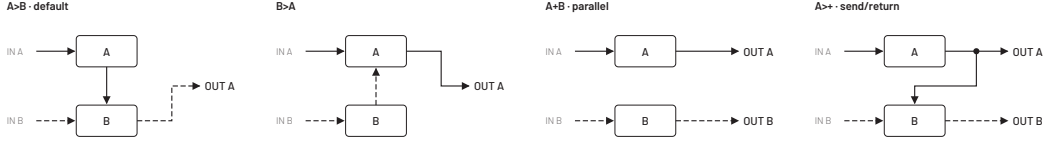
- **SHIFT + step 1–7** (on any page): pre-listen to a track, LED blinks.
- Pre-fader, also muted tracks. St2 becomes the cue bus automatically.
- **cmx**: cue ↔ main on St2, **cvl**: level.
- Needs cables in St1 (L and R); otherwise St1 plays on the St2 jacks.

6 > EFFECTS

6.1. SIGNAL FLOW



From the voice to the jacks, slot order A>B (dashed: slot B; ⇧ = SHIFT, K1 … K4 = parameter knobs). The fader acts on the sends only with **pre** off. FX3 sits on St1 (without a cable in St1 on the St2 jacks).



Order of the FX slots, **ord** on the FX1 page (chapter 6.2).

6.2. FX1 AND FX2

SHIFT + FX: FX1 (kit, slot A) › FX2 (project, slot B) › FX3 (master) › FX1. Both slots: **off drv rng cmp del**.

Display	Name	Description
typ / fx2	Type	effect of slot A / B
out	Audio out	output of the slot
d/w	Dry/Wet	mix; also SHIFT + knob 3 / 4

Setup page (turn past the last parameter):

Display	Name	Description
sya / syb	Sync A / B	delay time as a note value, 1/4 … 1/32T
ord	Order	FX1 page: routing of the slots
prs	Preset	click, choose, click: 17 factory presets (delay, drive, compressor, ring), **U1**–**U8** per project, **Save to U**n

ord	Routing
A>B	A into B, to the output of A (default). B is an insert: at 100 % wet the dry sound of A is missing.
B>A	B into A, to the output of A
A+B	parallel, each to its own output
A>+	send/return: A to its output and into B, B to its output

FX1 and **ord**: kit. FX2 and sync: global.

6.3. SENDS AND PRE-FADER

sna / **snb** (AEG page): sends into slot A / B. MIX **out** = **FXB**: 100 % into slot B.

Display	Name	Description
pre	Pre-fader sends	CONFIG, page 2. **off**: sends after the fader. **on**: before it, the fader sets only the dry signal (fader down = only the echo).

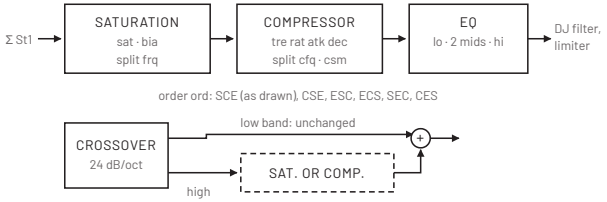
vol (MIX) always acts on both.

6.4. FX KNOBS

SHIFT + knob 1 / 2: input level FX1 / FX2 (−∞ … +12 dB, global); 3 / 4: mix. Not on LOAD/SAVE.

6.5. MASTER BUS FX3

Third slot at the end of **St1**: after FX1/FX2, before DJ filter and limiter. Saturation (S), compressor (C), EQ (E) in the order **ord**. In the kit.



Split: only the band above the crossover is saturated / compressed. **csm** = **sc**: the upper band only controls, everything is compressed.

FX3, PAGE 1

Display	Name	Description
sat	Saturation	0 = off; soft clipping up to +24 dB, level compensated
frq	Sat split	off or 40 Hz … 10 kHz
cfq	Comp split	off or 40 Hz … 10 kHz
csm	CFQ mode	**spl** band split, **sc** sidechain filter
tre	Threshold	–48 … 0 dB
rat	Ratio	0 = off, 1:1 … 20:1
atk	Attack	0.1 … 100 ms
dec	Release	**aut** = 1/16 of the tempo, else 20 … 1000 ms

FX3, PAGE 2: EQ

Display	Name	Description
lo	Low shelf	100 Hz, ±15 dB (64 = 0 dB = off)
f1 g1 q1	Mid 1	20 Hz … 16 kHz, ±15 dB, Q 0.3 … 8 (300 Hz, Q 1)
f2 g2 q2	Mid 2	as mid 1 (3 kHz)
hi	High shelf	8 kHz, ±15 dB

FX3, PAGE 3

Display	Name	Description
ord	Order	SCE (default), CSE, ESC, ECS, SEC, CES
amu	Auto gain	**off** · **fix** half the reduction of a 0 dBFS peak · **ada** average reduction (≈ 1.5 s)
gai	Gain	make-up 0 … +24 dB, in addition to **amu**
d/w	Dry/Wet	compressor mix (parallel)
bia	Sat bias	0 = symmetric; higher: even harmonics

NOTES

- **sat**, **rat**, **gai** at 0 and EQ at 0 dB: FX3 off.
- Band split: **amu**, **gai**, **d/w** only on the compressed band.
- **amu** = **ada**: **d/w** changes only the character, not the loudness.
- **bia** only with **sat** > 0; for a mix 20 … 50 with **frq** 150 … 300 Hz.

EDIT VIEW

Top: peaks St1 / St2 before FX3 (dBFS). Bottom: **byp** = off, **on-4A** = 4 dB reduction, St1 (**B**: St2 jacks).

7 > GLOBAL SETTINGS

Display	Name	SHIFT + CONFIG, page 2
lim	Limiter OE	on every output
ips	Instant pattern	chapter 4.4
mrv	Morph voices	kit morph per voice
pre	Pre-fader sends	chapter 6.3
aca	Accent amount	chapter 4.3

8 > MIDI

Message	Parameter (voice 1–6)
CC 1 · 2–5, 7, 8	**mrp** (global channel) · OSC **wav**
CC 122–127	LFO **mod**
NRPN 27–32 · 57–62	LFO **wav** · LFO **ofs**
NRPN 87–92 · 100–105	MIX **out** · OSC **sta**; **len**: 93, 94, 97
NRPN 98	engine OE: slot × 4 + engine, 24/30/36 + slot: modal/fm2/wt
NRPN 99 · 106–117	**chk** · FX1 OE

A > WHERE SETTINGS ARE STORED

Stored in	Settings
Kit	engine OE and its settings, **wav** (sample / wave number), **sta**, **len**, **chk**, **sna**, **snb**, **out**, LFO waveforms and **ofs**; FX1, **ord** of the FX slots; all of FX3; track 7, LFO 2, **osx**, **oct**
Pattern	step grid, flam values, accents **acc**, **oct** of the steps, start, trigger tracks, LFO start, steps with velocity 0
Global	FX2, **sya**, **syb**, the input levels (SHIFT + knob 1 / 2), **pre**, **ips**, **mrv**, **vsw**, **cmx**, **cvl**, **aca**, **rnd**, **rec**, **rst**, **lim** OE
Project	the INIT template for voices, the values of the morph kit, FX presets **U1**–**U8**
Flash	samples, waves and wavetables (imported from the card)
Not stored	**vmv**, **vmo**, **dj**, **res** OE

MORE INFORMATION

The full description, with more examples and the history of each version, is in the repository:

- **README**: overview of all features and the installation.
- **docs/SAMPLES.md**: guide to all additions.
- **docs/OVERVIEW.md** / **OVERVIEW.de.md**: one-page overview in English and German.
- **docs/ENGINES.md**: the voice engines OE.
- **docs/MOD_TARGETS.md**: the modulation targets of each voice.

github.com/shellbeat-jpg/lxr-desktop-sample

The XT firmware is free software under the GNU General Public License 3.0. It is based on the LXR firmware by Julian Schmidt (Sonic Potions) and Erica Synths and contains the modifications of OpenElixir-02 1.2 (Moritz Waser). Font: Barlow, SIL Open Font License.

B > MODULATION TARGETS

Targets of the LFOs, velocity and the step automation (parameter lock 1, 2). LFO and velocity **dst** show the name, the step page the voice and two letters (**1wa** = **wav** of voice 1, **0sr**: all voices). Targets the slot's engine lacks: **off**.

Page	Name	Edit view	Voices
Oscillator	**coa**	Oscilltr Coarse	1–6
	fin	Oscilltr Fine	1–6
	wav	Oscilltr Waveform	1–6
	pwm	Oscilltr PulsWdth	1–6
	noi	Noise Frequncy	4
	mix	Nois/Osc Mix	4
Amp EG	**atk**	Veloc EG Attack	1–6
	dec	Veloc EG Decay	1–5
	slp	Veloc EG Slope	1–6
	rpt	Veloc EG RepeatCt	4, 5
	d1	Veloc EG Dcy Clsd	6
	d2	Veloc EG Dcy Open	6
Modulation	**dec**	Mod EG Decay	1–4
	slp	Veloc EG Slope	1–4
	mod	PitchMod Amount	1–4
	dst	Velocity DstParam	1–6
	amt	Velocity Amount	1–6
	vol	Velocity Vol mod	1–6
FM	**amt**	FM Amount	1–3
	frq	FM Frequncy	1–3
	fwv	Oscilltr Waveform	1–3, 5, 6
FM	**mod**	Oscilltr Mode	1–3
	f1	Mod Osc Freqcy 1	5, 6
	f2	Mod Osc Freqcy 2	5, 6
	g1	Mod Osc Gain 1	5, 6
	g2	Mod Osc Gain 2	5, 6
Transient	**cwv**	Transnt Waveform	1–6
	vol	Transnt Volume	1–6
	frq	Transnt Frequncy	1–6
Filter	**frq**	Filter Frequncy	1–6
	res	Filter Resnance	1–6
	typ	Filter Type	1–6
	drv	Filter Overdriv	1–6
LFO	**frq**	LFO Frequncy	1–6
	snc	LFO ClockSnc	1–6
	mod	LFO Amount	1–6
	wav	LFO Waveform	1–6
	rtg	LFO Retriggr	1–6
	ofs	LFO Offset	1–6
Mix	**srt**	Voice SampleRt	0 (all), 1–6
	vol	Voice Volume	1–6
	pan	Voice Panning	1–6
	drv	Voice Overdriv	1–6
FX	**p1**	Fx Param 1	1–6
	p2	Fx Param 2	1–6
	p3	Fx Param 3	1–6
	p4	Fx Param 4	1–6

fwv = FM / mod oscillator, **cwv** = transient. A **wav** lock: 0–5 waveforms, 6–36 samples S1–S31, 37–44 waves W1–W8. All voices: docs/MOD_TARGETS.md.